

9.3 Sequential and partial sums of squares

How should the variation explained by a model be shared among its terms? A term is a block of columns representing one idea: a regressor, the coding columns of a factor, or an interaction. One answer enters the terms one at a time and credits each with what it adds to those already present. Another credits each term with what it adds when entered last. This section defines both, explains the three “types” of sums of squares that software prints, and identifies the hypothesis that each of them addresses.

9.3.1 Sequential sums of squares

Write the model matrix as a sequence of terms,

$$\mathbf{X} = [\mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_k],$$

where $\mathbf{X}_0$ is usually the intercept column **1**. Let $\mathbf{M}_j$ be the projection onto $\mathcal{C}(\mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_j)$, so that $\mathbf{M}_k = \mathbf{M}$.

Definition 9.3.1 — Sequential sums of squares

The **sequential** (or **Type I**) sum of squares of the term $\mathbf{X}_j$, for the order $\mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_k$, is

$$SS(\mathbf{X}_j \mid \mathbf{X}_0, \dots, \mathbf{X}_{j-1}) = \mathbf{y}^\top (\mathbf{M}_j - \mathbf{M}_{j-1}) \mathbf{y}, \quad j = 1, \dots, k,$$

with $d_j = \text{rank}(\mathbf{M}_j) - \text{rank}(\mathbf{M}_{j-1})$ degrees of freedom.

The sequential sums of squares are the analysis of variance decomposition of Theorem 9.1.1(e) for the chain $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X}_0, \mathbf{X}_1) \subseteq \dots \subseteq \mathcal{C}(\mathbf{X})$, after the first piece $\mathbf{M}_0$ has been set aside. They add up to the regression sum of squares $SS(\mathbf{X}_1, \dots, \mathbf{X}_k \mid \mathbf{X}_0)$, their degrees of freedom add up to $r - \text{rank}(\mathbf{X}_0)$, and under normality they are independent of each other and of the residual sum of squares (Theorem 4.4.4). A term linearly dependent on earlier ones receives zero degrees of freedom. The price is that each piece depends on the order: $\mathbf{X}_j$ is credited only with the part of its span orthogonal to *the terms that happen to precede it*. For the murder rates of [Section 6.5](#), poverty received 102.49 first and 16.37 last (Example (Order matters)).

The order dependence can be more surprising than “entering first gives more credit”. The listing computes $SS(\mathbf{x} \mid \mathbf{1}, S)$ for each of the three regressors and every set S of the other two.

Regressor	given 1 only	given 1 and one other (named)	given 1 and the remaining other (named)	given 1 and both others
poverty	102.49	19.69 (single)	125.45 (urban)	16.37
single	162.84	80.04 (poverty)	160.13 (urban)	51.06
urban	6.30	29.26 (poverty)	3.59 (single)	0.27

Adjusting for urbanization *increases* the sum of squares of poverty from 102.49 to 125.45, and adjusting for poverty more than quadruples that of urbanization. Urban states have somewhat less poverty but more murder, so holding one fixed sharpens the association of the other; such a

regressor is called a **suppressor** (Exercise (B1)). A sequential sum of squares is not a property of a regressor, and not even a monotone function of the set adjusted for.

```
from itertools import combinations
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statcrimeload_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
n = len(y)
one = np.ones(n)
cols = {"poverty": data["poverty"].to_numpy(), "single": data["single"].to_numpy(),
        "urban": data["urban"].to_numpy()}

def sse(Z):
    b, *_ = np.linalg.lstsq(Z, y, rcond=None)
    e = y - Z @ b
    return e @ e

table = {}
for name, x in cols.items():
    others = [k for k in cols if k != name]
    for size in range(3):
        for S in combinations(others, size):
            base = np.column_stack([one] + [cols[k] for k in S])
            table[name, S] = sse(base) - sse(np.column_stack([base, x]))
            print(f"SS({name} | 1{''.join(', ' + k for k in S)}) = {table[name, S]
                  }:7.2f}")
```

[Run this code online](#)

9.3.2 Partial sums of squares: Types II and III

The second answer credits a term with what it adds last. For models whose terms are factors and their interactions, “last” needs care, because a main effect is part of every interaction that contains it. Say that a term $\mathbf{X}_i$ **contains** a term $\mathbf{X}_j$ if the factors of $\mathbf{X}_j$ are a proper subset of the factors of $\mathbf{X}_i$. For example the interaction AB contains the main effects A and B .

Definition 9.3.2 — Partial sums of squares; Types II and III

For a model with terms $\mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_k$, write $\mathbf{X}_{(-j)}$ for $\mathbf{X}$ with the columns of $\mathbf{X}_j$ removed.

- (a) The **partial** (or **last-in**) sum of squares of $\mathbf{X}_j$ is $SS(\mathbf{X}_j \mid \mathbf{X}_{(-j)}) = \mathbf{y}^\top (\mathbf{M} - \mathbf{M}_{(-j)}) \mathbf{y}$, where $\mathbf{M}_{(-j)}$ projects onto $\mathcal{C}(\mathbf{X}_{(-j)})$.
- (b) The **Type II** sum of squares of $\mathbf{X}_j$ is its extra sum of squares given all terms that do not contain it: $SS(\mathbf{X}_j \mid \mathbf{X}_i : \mathbf{X}_i \text{ does not contain } \mathbf{X}_j, i \neq j)$.
- (c) The **Type III** sum of squares of $\mathbf{X}_j$ is its partial sum of squares (a) computed when every factor is coded by sum-to-zero contrast columns and every interaction by the products of the columns of its factors.

When no term contains another, as in a regression on numerical regressors without interactions, the three coincide, and they equal the sequential sum of squares of $\mathbf{X}_j$ when it is entered last.

They also coincide for a term that is not contained in any other, such as the highest-order interaction. They differ for main effects in models with interactions.

Partial sums of squares are not a decomposition. The subspaces $\mathcal{C}(\mathbf{X}) \ominus \mathcal{C}(\mathbf{X}_{(-j)})$ are generally not orthogonal, so the partial sums of squares can add up to more or less than the regression sum of squares ([Exercise \(C1\)](#)), and they need not be independent of each other, though each is independent of the residual sum of squares under normality.

Why does the definition of Type III name a coding? Because in models with interactions the partial sum of squares of a main effect depends on the coding. Deleting the columns of A while keeping those of AB leaves the reduced space $\mathcal{C}(\mathbf{1}, \mathbf{X}_B, \mathbf{X}_{AB})$, which depends on which columns represent AB , although the full space does not. Different codings give different reduced models and so test different hypotheses.

9.3.3 What each sum of squares tests

A sum of squares $\mathbf{y}^\top \mathbf{P} \mathbf{y}$ with $\mathbf{P}$ a projection of rank d has expectation

$$E(\mathbf{Y}^\top \mathbf{P} \mathbf{Y}) = \sigma^2 d + \|\mathbf{P} \boldsymbol{\mu}\|^2, \quad \boldsymbol{\mu} = E(\mathbf{Y}),$$

by Theorem 2.3.1 with $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$. So the hypothesis a sum of squares addresses is $\mathbf{P} \boldsymbol{\mu} = \mathbf{0}$: *the mean has no component in the subspace onto which it projects*. Under normality this is exactly the condition for its F ratio to be central ([Section 9.6](#)). We translate it into statements about parameters.

Proposition 9.3.1 — Hypotheses of sequential and partial sums of squares

Let $\mathbf{X} = [\mathbf{X}_0, \dots, \mathbf{X}_k]$ have full column rank and $\boldsymbol{\mu} = \sum_l \mathbf{X}_l \beta_l$.

- (a) The partial sum of squares of $\mathbf{X}_j$ addresses $\beta_j = \mathbf{0}$: $(\mathbf{M} - \mathbf{M}_{(-j)}) \boldsymbol{\mu} = \mathbf{0}$ iff $\beta_j = \mathbf{0}$.
- (b) The sequential sum of squares of $\mathbf{X}_j$ addresses

$$\beta_j + (\tilde{\mathbf{X}}_j^\top \tilde{\mathbf{X}}_j)^{-1} \tilde{\mathbf{X}}_j^\top \sum_{l>j} \mathbf{X}_l \beta_l = \mathbf{0}, \quad \tilde{\mathbf{X}}_j = (\mathbf{I} - \mathbf{M}_{j-1}) \mathbf{X}_j.$$

This is $\beta_j = \mathbf{0}$ whenever the later terms have zero coefficients, or whenever $\tilde{\mathbf{X}}_j^\top \mathbf{X}_l = \mathbf{0}$ for all $l > j$.

Proof.

- (a) For $l \neq j$, $\mathbf{X}_l \beta_l$ lies in $\mathcal{C}(\mathbf{X}_{(-j)}) \subseteq \mathcal{C}(\mathbf{X})$, so $(\mathbf{M} - \mathbf{M}_{(-j)}) \mathbf{X}_l \beta_l = \mathbf{0}$. Let $\tilde{\mathbf{X}}_j = (\mathbf{I} - \mathbf{M}_{(-j)}) \mathbf{X}_j$. By Lemma 9.2.1, $\mathbf{M} - \mathbf{M}_{(-j)}$ projects onto $\mathcal{C}(\tilde{\mathbf{X}}_j)$, so $(\mathbf{M} - \mathbf{M}_{(-j)}) \mathbf{X}_j \beta_j$ is the projection of $\mathbf{X}_j \beta_j$ onto that space. Since $\mathbf{X}_j \beta_j - \tilde{\mathbf{X}}_j \beta_j = \mathbf{M}_{(-j)} \mathbf{X}_j \beta_j$ is orthogonal to $\mathcal{C}(\tilde{\mathbf{X}}_j)$, the projection is $\tilde{\mathbf{X}}_j \beta_j$. The columns of $\tilde{\mathbf{X}}_j$ are independent (Lemma 6.6.1(b)), so it vanishes iff $\beta_j = \mathbf{0}$.
- (b) Terms with $l < j$ are annihilated by $\mathbf{M}_j - \mathbf{M}_{j-1}$, which projects onto $\mathcal{C}(\tilde{\mathbf{X}}_j)$ by Lemma 9.2.1. As in (a), $\mathbf{X}_j \beta_j$ projects to $\tilde{\mathbf{X}}_j \beta_j$. A later term projects to $\tilde{\mathbf{X}}_j (\tilde{\mathbf{X}}_j^\top \tilde{\mathbf{X}}_j)^{-1} \tilde{\mathbf{X}}_j^\top \mathbf{X}_l \beta_l$ by Theorem 6.3.7. Adding, $(\mathbf{M}_j - \mathbf{M}_{j-1}) \boldsymbol{\mu} = \tilde{\mathbf{X}}_j \mathbf{v}$, where $\mathbf{v}$ is the vector in the display, and this is zero iff $\mathbf{v} = \mathbf{0}$ because $\tilde{\mathbf{X}}_j$ has independent

columns.

■

So a sequential sum of squares answers the question “does $\mathbf{X}_j$ matter?” only if the terms fitted after it do not matter, or are orthogonal to it after adjusting for the terms before it. Otherwise the effects of later terms leak into it, through the coefficients $(\tilde{\mathbf{X}}_j^\top \tilde{\mathbf{X}}_j)^{-1} \tilde{\mathbf{X}}_j^\top \mathbf{X}_l$ of the regressions of later terms on $\tilde{\mathbf{X}}_j$.

For factors, the natural parameters are the cell means. Consider a two-way layout with factor A at a levels and B at b levels, $n_{ij} \geq 1$ observations in cell (i, j) , row totals $n_{i\cdot}$, and cell means μ_{ij} . The full model with interaction has model space $\mathcal{C}(\mathbf{X}) =$ all vectors that are constant within cells, of dimension ab . Define the **weighted** and **unweighted** row means

$$\bar{\mu}_i^w = \frac{1}{n_{i\cdot}} \sum_j n_{ij} \mu_{ij}, \quad \bar{\mu}_{i\cdot} = \frac{1}{b} \sum_j \mu_{ij}.$$

Theorem 9.3.2 — What the sums of squares for a main effect test

In the two-way layout with all $n_{ij} \geq 1$:

- (a) the sequential sum of squares $\text{SS}(A \mid \mu)$ of A entered first addresses $\bar{\mu}_1^w = \cdots = \bar{\mu}_a^w$;
- (b) the Type III sum of squares of A addresses $\bar{\mu}_{1\cdot} = \cdots = \bar{\mu}_{a\cdot}$;
- (c) the partial sum of squares of the columns of A under reference-level coding (indicators of levels $2, \dots, a$ of A and $2, \dots, b$ of B , and their products for AB) addresses $\mu_{11} = \mu_{21} = \cdots = \mu_{a1}$, the equality of the A means at the first level of B ;
- (d) the Type II sum of squares $\text{SS}(A \mid \mu, B)$ addresses a hypothesis that, when the cell means are additive ($\mu_{ij} = \alpha_i + \beta_j$), is equivalent to $\alpha_1 = \cdots = \alpha_a$.

Each sum of squares equals the sum of squares Eq. 9.2.1 of the hypothesis it addresses, in the cell-means model.

Proof. Let $\mathbf{P}_0$, $\mathbf{P}_A$, $\mathbf{P}_B$ and $\mathbf{P}_{A+B}$ be the projections onto the constant vectors, the vectors constant within rows, those constant within columns, and their sum. Each hypothesis below defines a subspace $\mathcal{S}_0$ of $\mathcal{C}(\mathbf{X})$, and each sum of squares is $\|\mathbf{P}\mathbf{y}\|^2$ with $\mathbf{P}$ the projection onto $\mathcal{C}(\mathbf{X}) \ominus \mathcal{S}_0$. Then $\mathbf{P}\boldsymbol{\mu} = \mathbf{0}$ iff $\boldsymbol{\mu} \in \mathcal{S}_0$, and the sum of squares equals $\text{SSE}_0 - \text{SSE}$ for $\mathcal{S}_0$, which is the sum of squares of the hypothesis by Theorem 9.2.4.

- (a) $\mathbf{P}_A$ replaces each entry of a vector by the average of its row, so $\mathbf{P}_A\boldsymbol{\mu}$ has entry $\bar{\mu}_i^w$ in row i , and $\mathbf{P}_0\boldsymbol{\mu}$ has every entry equal to the overall weighted mean $\sum_i n_{i\cdot} \bar{\mu}_i^w / n$. So $(\mathbf{P}_A - \mathbf{P}_0)\boldsymbol{\mu} = \mathbf{0}$ iff all $\bar{\mu}_i^w$ equal their weighted average, that is, iff they are all equal. The relevant $\mathcal{S}_0$ is $\{\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X}) : (\mathbf{P}_A - \mathbf{P}_0)\boldsymbol{\mu} = \mathbf{0}\}$, whose orthogonal complement in $\mathcal{C}(\mathbf{X})$ is $\mathcal{C}(\mathbf{P}_A - \mathbf{P}_0)$, because $\mathcal{C}(\mathbf{P}_A - \mathbf{P}_0) \subseteq \mathcal{C}(\mathbf{X})$.
- (b) Any $a \times b$ array of cell means decomposes uniquely as

$$\mu_{ij} = \bar{\mu}_{\cdot\cdot} + (\bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot\cdot}) + (\bar{\mu}_{\cdot j} - \bar{\mu}_{\cdot\cdot}) + (\mu_{ij} - \bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot j} + \bar{\mu}_{\cdot\cdot}),$$

where bars denote unweighted averages. The four arrays lie in the mutually orthogonal subspaces (for the unweighted inner product on arrays) of constants, row

effects summing to zero, column effects summing to zero, and arrays with all row and column sums zero, of dimensions $1, a-1, b-1, (a-1)(b-1)$. Sum-to-zero coding spans exactly these: the columns $\mathbf{u}_i - \mathbf{u}_1$ ($i = 2, \dots, a$), where $\mathbf{u}_i$ indicates row i , give all row effects α_i with $\sum_i \alpha_i = 0$; similarly for B ; and the $(a-1)(b-1)$ products are linearly independent arrays with zero row and column sums, so they span that subspace. Deleting the columns of A removes exactly the row-effect component, so the reduced model is $\mathcal{S}_0 = \{\boldsymbol{\mu} : \bar{\mu}_{i.} = \bar{\mu}_{..} \text{ for all } i\}$.

- (c) With reference coding, every cell-mean array is uniquely $\mu_{ij} = m + \alpha_i + \beta_j + \gamma_{ij}$ with $\alpha_1 = \beta_1 = 0$ and $\gamma_{1j} = \gamma_{i1} = 0$; explicitly $m = \mu_{11}$, $\alpha_i = \mu_{i1} - \mu_{11}$, $\beta_j = \mu_{1j} - \mu_{11}$ and $\gamma_{ij} = \mu_{ij} - \mu_{i1} - \mu_{1j} + \mu_{11}$. Deleting the columns of A forces $\alpha_i = 0$, that is, $\mu_{i1} = \mu_{11}$ for all i .
- (d) The Type II sum of squares is $\mathbf{y}^\top (\mathbf{P}_{A+B} - \mathbf{P}_B) \mathbf{y}$. If $\boldsymbol{\mu}$ is additive, then $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{P}_{A+B})$ and $(\mathbf{P}_{A+B} - \mathbf{P}_B)\boldsymbol{\mu} = \boldsymbol{\mu} - \mathbf{P}_B\boldsymbol{\mu}$, which is zero iff $\boldsymbol{\mu}$ is constant within columns, that is, iff α_i does not depend on i .

■

These are four different questions. Part (a) averages over B with the design's weights, so if B matters and the rows have different mixes of B levels, it can detect “an effect of A ” that is entirely an effect of B (Exercise B3). Part (b) averages with equal weights. Part (c) compares the rows at one level of B : a simple effect, not a main effect, and one that changes with the reference level. Part (d) is the natural test of A without interaction; with interaction its hypothesis has weights that depend on all the n_{ij} .

Warning (“Type III” with the default coding). Many regression programs code factors by reference levels unless told otherwise. Asking them for last-in sums of squares of main effects in a model with interactions then produces the hypotheses of Theorem 9.3.2(c), which concern a single level of the other factor. They are not Type III sums of squares, although they are sometimes reported under that name. Use sum-to-zero coding, or better, write down the hypothesis and use Eq. 9.2.1.

9.3.4 An unbalanced example

Example. (Party, education and political self-placement)

In the 1996 American National Election Study, 944 respondents placed themselves on a seven-point scale from extremely liberal (1) to extremely conservative (7). Classify them by party identification (Democrat, Independent, Republican, counting those who only lean to a party as Independents) and by education (high school or less, college, graduate study). The cell counts and means are:

	high school	college	graduate
Democrat	146, 3.97	108, 3.43	126, 2.92
Independent	80, 4.39	64, 4.28	95, 4.09
Republican	87, 5.40	105, 5.36	133, 5.40

The design is unbalanced, and education is associated with party: 38% of the Democrats but only 27% of the Republicans have at most a high school education. The pattern of means shows an interaction. Among Democrats self-placement moves left with education by more than a scale point, while among Republicans it does not move at all. The sums of squares for the terms are:

Term	df	Type I, party first	Type I, education first	Type II	Type III	reference coding
party	2	647.46	674.42	674.42	650.41	112.27
education	2	43.26	16.31	43.26	32.03	75.12
interaction	4	35.81	35.81	35.81	35.81	35.81
residual	935	1224.62	1224.62	1224.62	1224.62	1224.62

The two Type I columns each add up to the model sum of squares 726.54, up to rounding. The other columns do not: the Type II entries add up to more. Education receives 16.31 when entered first and 43.26 after party. Since the least educated group contains the most Democrats, who are the most liberal, the marginal comparison mixes the leftward drift of educated Democrats with the partisan composition of each education group, and the two partly cancel. The Type III sum of squares 32.03 tests whether the unweighted averages over the three parties of the education means are equal. The last column, computed with reference coding, answers different questions: 75.12 for education measures the education gradient among *Democrats only* (the first party level), which is the steepest, and 112.27 for party compares the parties among respondents with a high school education only. The interaction row is the same in every column because the interaction is the highest-order term.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
d = sm.datasets.anes96.load_pandas().data
party = pd.cut(d["PID"], [-1, 1.5, 4.5, 6], labels=["Dem", "Ind", "Rep"]).rename("party")
educ = pd.cut(d["educ"], [0, 3.5, 5.5, 7], labels=["HS", "College", "Graduate"]).rename("educ")
y = d["selfLR"].to_numpy()
n = len(y)
print(pd.crosstab(party, educ))
print(d.groupby([party, educ], observed=True)["selfLR"].mean().unstack().round(3))
```

[Run this code online](#)

```
def proj(*blocks):
    """Orthogonal projection onto the span of the given column blocks."""
    Z = np.column_stack(blocks)
    U, s, _ = np.linalg.svd(Z, full_matrices=False)
    U = U[:, s > 1e-10 * s[0]] # orthonormal basis of C(Z), any
    rank                        rank
    return U @ U.T

one = np.ones((n, 1))
A = pd.get_dummies(party).to_numpy(float) # indicators of party
B = pd.get_dummies(educ).to_numpy(float)   # indicators of education
```

```

AB = np.column_stack([A[:, [i]] * B for i in range(3)]) # the nine cells
P0, PA, PB = proj(one), proj(one, A), proj(one, B)
PAB, M = proj(one, A, B), proj(AB) # additive model, cell-means
model
I = np.eye(n)

def ss(P):
    return y @ P @ y

type1_AB = {"party": ss(PA - P0), "educ": ss(PAB - PA), "inter": ss(M - PAB)}
type1_BA = {"educ": ss(PB - P0), "party": ss(PAB - PB), "inter": ss(M - PAB)}
type2 = {"party": ss(PAB - PB), "educ": ss(PAB - PA), "inter": ss(M - PAB)}
sse = ss(I - M)

```

[Run this code online](#)

```

def coded(labels, kind):
    """Columns for a factor: sum-to-zero ("sum") or reference-level ("treatment")
    coding."""
    D = pd.get_dummies(labels).to_numpy(float)
    return D[:, 1:] - D[:, [0]] if kind == "sum" else D[:, 1:]

type3 = {}
for kind in ["sum", "treatment"]:
    Ca, Cb = coded(party, kind), coded(educ, kind)
    Cab = np.column_stack([Ca[:, [i]] * Cb for i in range(2)])
    full = proj(one, Ca, Cb, Cab) # always the cell-means space
    type3[kind] = {"party": ss(full - proj(one, Cb, Cab)),
                  "educ": ss(full - proj(one, Ca, Cab))}
    assert np.allclose(full, M)
for name, table in [("Type I, party first", type1_AB), ("Type I, education first",
type1_BA),
                    ("Type II", type2), ("Type III, sum coding", type3["sum"]),
                    ("Type III, treatment coding", type3["treatment"])] :
    print(f"{name:30s}", {k: round(float(v), 2) for k, v in table.items()})

```

[Run this code online](#)

The script behind these cells also checks every entry against statsmodels, and it confirms [Theorem 9.3.2](#) numerically: the party entries of the first, fourth and fifth columns equal Eq. 9.2.1 for the hypotheses of equal weighted row means, equal unweighted row means, and equal means in the high-school column.

9.3.5 Which one to use

There is no single right sum of squares for a main effect in an unbalanced design with interaction, because there is no single main-effect hypothesis. Decide the hypothesis first, from the four above, and compute its sum of squares with Eq. 9.2.1. If A and B interact, a single “main effect of A ” summarizes a heterogeneous comparison with arbitrary weights; Nelder (1977) argued that such tests are rarely meaningful, and Type II sums of squares respect this marginality by adjusting A for B but not for AB . Without interaction the Type II and Type III hypotheses coincide, and Type II, which does not adjust for the interaction columns, generally gives the more powerful test. In regression without interactions all partial types agree; the distinction matters only for terms contained in other terms.

Chapter 17 treats empty cells, where even Type III hypotheses may be nonestimable (Exercise (B4), Theorem 17.4.1).

9.3.6 Exercises

9.3.6.1 A. Check your understanding

Exercise A1.

For a two-way model with terms A , B and AB , write each of the Type I sums of squares (both orders), the Type II sums of squares and the partial sums of squares in the notation $SS(\cdot | \cdot)$. Which of them are equal for every data set, whatever the design?

Solution. Type I, A first: $SS(A | \mu)$, $SS(B | \mu, A)$, $SS(AB | \mu, A, B)$. Type I, B first: $SS(B | \mu)$, $SS(A | \mu, B)$, $SS(AB | \mu, A, B)$. Type II: $SS(A | \mu, B)$, $SS(B | \mu, A)$, $SS(AB | \mu, A, B)$. Partial: $SS(A | \mu, B, AB)$, $SS(B | \mu, A, AB)$, $SS(AB | \mu, A, B)$, where the first two depend on the coding. Always equal: the interaction entries of all four lists; the Type II entry for A and the Type I entry for A with B first; the Type II entry for B and the Type I entry for B with A first.

9.3.6.2 B. Practice

Exercise B1.

Let $\mathbf{X} = [\mathbf{1}, \mathbf{x}_1, \mathbf{x}_2]$, and let r_{y1} , r_{y2} and r_{12} be the sample correlations among $\mathbf{y}$, $\mathbf{x}_1$ and $\mathbf{x}_2$, with $|r_{12}| < 1$. Show that

$$\frac{SS(\mathbf{x}_2 | \mathbf{1})}{SST} = r_{y2}^2, \quad \frac{SS(\mathbf{x}_2 | \mathbf{1}, \mathbf{x}_1)}{SST} = \frac{(r_{y2} - r_{y1}r_{12})^2}{1 - r_{12}^2}.$$

Show that adjusting for $\mathbf{x}_1$ increases the sum of squares of $\mathbf{x}_2$ whenever $r_{y2} = 0$ and $r_{y1}r_{12} \neq 0$, and find the condition for it in general.

Solution. Work in $\mathbf{1}^\perp$ with the centred vectors $\mathbf{u} = \mathbf{y} - \bar{y}\mathbf{1}$, $\mathbf{v}_1$, $\mathbf{v}_2$, scaled to unit length, so that inner products are correlations and $SST = \|\mathbf{y} - \bar{y}\mathbf{1}\|^2$. By Lemma 9.2.1, $SS(\mathbf{x}_2 | \mathbf{1}) = SST(\mathbf{u}^\top \mathbf{v}_2)^2 = SST r_{y2}^2$. Adjusting for $\mathbf{x}_1$ replaces $\mathbf{v}_2$ by $\mathbf{w} = \mathbf{v}_2 - r_{12}\mathbf{v}_1$, with $\|\mathbf{w}\|^2 = 1 - r_{12}^2$ and $\mathbf{u}^\top \mathbf{w} = r_{y2} - r_{y1}r_{12}$, so $SS(\mathbf{x}_2 | \mathbf{1}, \mathbf{x}_1) = SST(\mathbf{u}^\top \mathbf{w})^2 / \|\mathbf{w}\|^2$. If $r_{y2} = 0$ the first ratio is zero and the second is $r_{y1}^2 r_{12}^2 / (1 - r_{12}^2) > 0$. In general the sum of squares increases iff $(r_{y2} - r_{y1}r_{12})^2 > r_{y2}^2(1 - r_{12}^2)$.

Exercise B2.

Show that the hypotheses of Theorem 9.3.2(a) and (b) are the same subspace of cell means iff $n_{ij} = n_{i.}/b$ for all i, j , that is, iff within every row all cells have the same count. Show that under proportional frequencies, $n_{ij} = n_{i.}n_{.j}/n$ (Theorem 9.4.3), the hypothesis (a) compares row means weighted by the column proportions $n_{.j}/n$, the same weights in every row.

Solution. Write $w_{ij} = n_{ij}/n_{i\cdot}$, so $\sum_j w_{ij} = 1$, and let $f_i(\boldsymbol{\mu}) = \sum_j w_{ij}\mu_{ij}$ and $g_i(\boldsymbol{\mu}) = \sum_j \mu_{ij}/b$. Hypothesis (a) is the null space of the linear functionals $\sum_i c_i f_i$ with $\sum_i c_i = 0$, and (b) that of the functionals $\sum_i d_i g_i$ with $\sum_i d_i = 0$. Two subspaces defined as null spaces coincide iff the spaces of functionals coincide. The functional $f_i - f_1$ has coefficient w_{ij} on cell (i, j) and $-w_{1j}$ on cell $(1, j)$. If it equals some $\sum_i d_i g_i$, whose coefficients are constant across each row, then w_{ij} does not depend on j , so $w_{ij} = 1/b$; likewise $w_{1j} = 1/b$. Hence coincidence forces $n_{ij} = n_{i\cdot}/b$. Conversely, if all $w_{ij} = 1/b$, then $f_i = g_i$. Under proportional frequencies $w_{ij} = n_{\cdot j}/n$, which does not depend on i , so (a) compares $\sum_j (n_{\cdot j}/n)\mu_{ij}$ across rows.

Exercise B3.

Take $a = b = 2$, counts $n_{11} = n_{22} = 9$ and $n_{12} = n_{21} = 1$, and additive cell means $\mu_{ij} = \beta_j$ with $\beta_1 = 0$ and $\beta_2 = 1$, so that A has no effect. Compute the weighted row means $\bar{\mu}_i^w$ and show that the hypothesis addressed by the Type I sum of squares of A entered first is false. What do the Type II and Type III hypotheses say here?

Exercise B4.

Suppose that in a 2×2 layout the cell $(2, 2)$ is empty. Show that $\bar{\mu}_{1\cdot} - \bar{\mu}_{2\cdot}$ is not estimable. Compute what the last-in sum of squares of A under sum-to-zero coding tests in this case. *Hint:* with a cell missing, the interaction columns restricted to the observed cells are no longer independent of the main-effect columns.

9.3.6.3 C. Going deeper

Exercise C1.

With the notation of [Exercise \(B1\)](#), show that

$$\frac{SS(\mathbf{x}_1 \mid \mathbf{1}, \mathbf{x}_2) + SS(\mathbf{x}_2 \mid \mathbf{1}, \mathbf{x}_1) - SSR}{SST} = \frac{r_{12}[r_{12}(r_{y1}^2 + r_{y2}^2) - 2r_{y1}r_{y2}]}{1 - r_{12}^2}.$$

Deduce that the partial sums of squares add up to the regression sum of squares when $r_{12} = 0$, that they add up to less when $r_{y1} = r_{y2} \neq 0$ and $0 < r_{12} < 1$, and give correlations for which they add up to more.